



JS MARINE DEFENCE

8B Parkstone Bay Marina

Turks Lane

Poole, Dorset

UK

info@JSMDDefence.com | 01202 738004

JSMD R5 – Technical Specification – Dynamic Target USV



Overview	
Vessel Type	Uncrewed Surface Vessel
Model	JSMD R5
Classification	Diesel USV Dynamic Training Target
Length	19ft / 5.7m (incl outdrive)
Beam	2.1m
Draft	0.4m (Hull only) / 0.8m (Drive down)
Displacement	1,200kg
Max Speed	35Knts Remotely (The hull is capable of 70knts)
Tank Capacity	160 litres

Hull and Structure	
1 x 5.7M JSMD R5 Hull with integrated closed cell foam.	Layup to ISO 12215-5 (Cat. C). Towing-point reinforcement per BV NR546. Up to Sea State 4.
1 x R5 GRP Mast Pack with 4 x integrated camera mounts and aerals mounts. Thermal camera / FLIR mount integrated.	
Zipwake CP-S with 2 x 300S interceptors	

Powertrain / Propulsion	
1 x Yanmar 4LV250Z Inboard Diesel. 250hp.	Variable HP available 150-250 dependant on max performance requirement.
1 x Yanmar ZT370 Outdrive	

Fuel System	
Tank Capacity 103L Fuel tank	(Placeholder, pending duty-cycle finalisation)
Roto-moulded polyethylene tank (ISO 21487), drop-in mounted within a banded GRP bay lined with absorbent spillage matting.	
Capacitive tank-level sensor reporting to telemetry, with low-fuel alarm.	
Primary water-separating filter with secondary engine-mounted filter	

Autonomy, Control and Navigation	
Autonomy Controller	WAGO Programable Logic Controller. CanBus, Digital I/O, ethernet and serial system for full vessel control. Fanless rugged PC system. Autonomy brain and system controller.
GNSS	Garmin 24XD GPS
AIS	Emtrak B920 AIS (Class B)
Compass	Maretron Solid State Compass
NMEA	NMEA2000 backbone
Camera	4 x External IP camera system (fixed) HikVision camera day/night mode. Can be upgraded to an AXIS IP Camera system. 1 x Internal IP camera on engine bay.
Kill switch	433mhz or 868mhz for direct engine kill
Wiring loom	Quick change wiring loom, protected and sleeved with Deutsch plug connectors.
Mounting	Pelicans hard shell case, shock mounted.

Communication and Networking	
Primary communication device	Starlink Mini Terminal 1
Secondary communication device	Starlink Mini Terminal 2
Out of bandwidth / emergency communication device	RF Kill switch. Independent hardware kill switch direct to engine over secure RF channel (868mhz) up to 10km range dependent on antenna height.
Network Switch	Mikro Tik network switch with POE. VPN network or P2P. Load bearing capability or fall-back comms. Multiple configuration modes.

Ancillaries	
Navigation Lights	(MH, Stern, Port, Stbd)
Bilge Pumps - 3,700–4,500 LPH	2x Attwood S1200
Steering System	CMP Octopus
Radar reflector	Interchangeable towing-point fitting accepts a radar reflector pole for enhanced radar cross-section and additional payload mounting interface.

Fire Control	
Remotely activated Fire Suppression System	STAT-X 30E
Fire detector	Smoke activated fire detector
Temperature Sensor	Heat compartment sensor

Ballistic Protection	
Bespoke Ballistic protection aligned to EN. 1063, N.I.J 0108.01 and STANAG 4569	

Documentation	
Vessel Administration Documentation	Operations Manual Technical Manual Maintenance Manual Software Operation Manual Emergency Stop Procedure Pre-Departure Checks Procedure Maintenance Interval Schedule
Drawings	General Arrangement Drawing Mast Arrangement Drawing Electrical Layout Drawing Plumbing Layout Drawing IT Network Drawing
Test Documentation	FAT (Factory Acceptance Test) SAT (Site Acceptance Test) Commissioning Report
Risk Assessments	Launch & Recovery Risk Assessment Towing Operations Risk Assessment Collision & Navigation Risk Assessment
Other RA / SOP's on request	

OPTIONAL: Remote Operating Centre	
Remote control station	Getac S510AD Laptop Computer Windows 11

	LAN, HDMI, 16GB RAM, 15.6", USB-C charger
Monitor	2x EIZO FlexScan 24" monitors Built in KVM for two monitors, flicker free, reduced eye strain for long term use
Software	CET BlueHelm v2.0 Core Package Lifetime License

OPTIONAL: Handheld Controller	
Handheld remote controller	GETAC UX10 Tablet Windows 11 Requires LTE Sim card for operation or WAN access from local onshore network such as WIFI or LAN. (Hand joystick system built in)
Software	CET BlueHelm v2.0 Core Package Lifetime License